Flowing Continua in 3D

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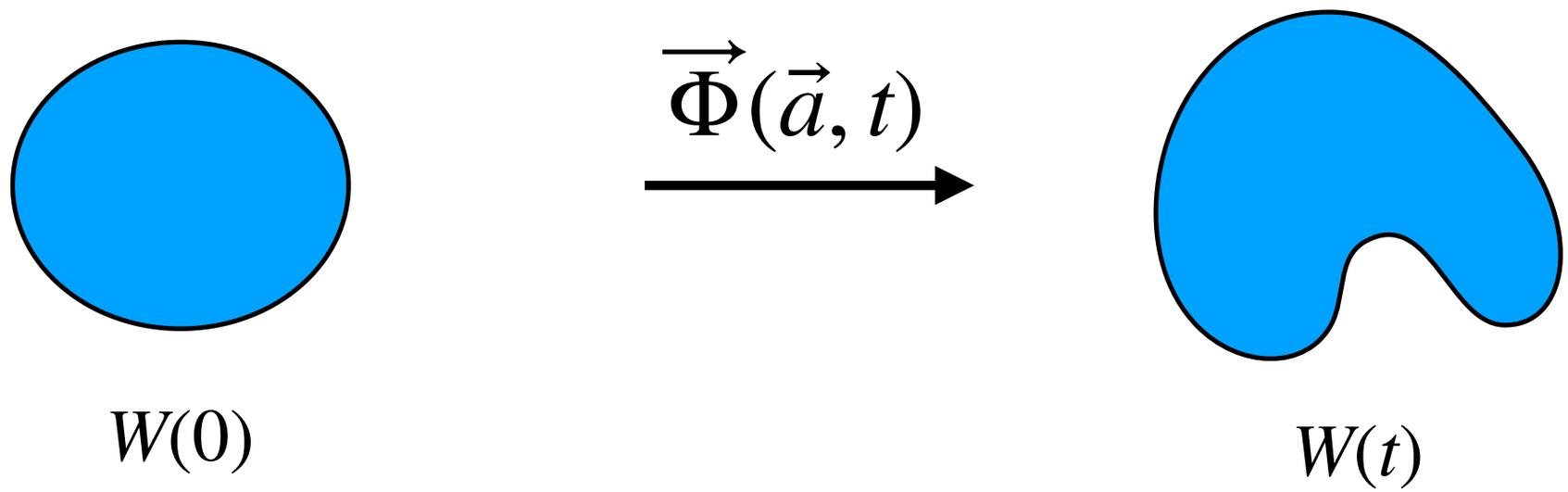
As in 1D, $W(t)$ is called the material volume



- The mapping is smooth and invertible.

- It also identifies a collection of fluid particles in the sense of Euler.

$$F(t) = \iiint_{W(t)} f(\vec{x}, t) dV$$

Small enough to be treated as points, large enough to have measurable properties like temperature.

- Calculating $\dfrac{dF}{dt}$ is now a much more complicated task and hence requires a separate theorem.

- The result in question is called the Reynolds Transport Theorem and states:

The Reynolds Transport Theorem

Local change of volume

$$\frac{dF}{dt} = \iiint_{W(t)} \left[\frac{Df}{Dt} + f \nabla \cdot \vec{u} \right] dV$$

Rate of change following a fluid particle

- The Reynolds Transport Theorem, along with the Dubois-Reymond Lemma, gives the Conservation of Mass right away:

$$\frac{dM}{dt} = 0 \implies \boxed{\frac{D\rho}{Dt}} + \boxed{\rho \nabla \cdot \vec{u}} = 0$$

When the density is constant this reduces to a conservation of volume statement:

The Reynolds Transport Theorem

$$\frac{dM}{dt} = 0 \implies \boxed{\frac{D\rho}{Dt}} + \boxed{\rho \nabla \cdot \vec{u}} = 0$$

Extra assumption:

$$\rho = \rho_0 \implies \frac{D\rho}{Dt} = \frac{D\rho_0}{Dt} = 0$$

$$\rho_0 \vec{\nabla} \cdot \vec{u} = 0$$

When the density is constant this reduces to a conservation of volume statement:

$$\boxed{\nabla \cdot \vec{u} = 0}$$

OK, what about if we have sources and sinks?

$$\frac{dM}{dt} = S(t)$$

With sources and sinks the rate of change of total mass is balanced by the total of sources and sinks

Apply the Transport Theorem to the LHS, and integrate the sum of sources and sinks over the material volume on the RHS

$$\iiint_{W(t)} \left[\frac{D\rho}{Dt} + \rho \nabla \cdot \vec{u} \right] dV = \iiint_{W(t)} S(\vec{x}, t) dV$$

$$\iiint_{W(t)} \left[\frac{D\rho}{Dt} + \rho \nabla \cdot \vec{u} - S(\vec{x}, t) \right] dV = 0$$

Rearrange and apply the Dubois Reymond Lemma to get the PDE, which has the source/sink on the RHS

$$\boxed{\frac{D\rho}{Dt}} + \boxed{\rho \nabla \cdot \vec{u}} = \boxed{S(\vec{x}, t)}$$

- How does one prove the Reynolds Transport Theorem?

- It is really a game of two steps; one conceptual and one technical.

- The technical Lemma, as is the case many times, does all the work and is the more slippery proof (I can provide references).

Lemma: If Φ is twice continuously differentiable and invertible (which we have already assumed) then

1. $J > 0$

   This follows from the definition of Φ

2. The time derivative is given by

$$\frac{\partial J}{\partial t} = (\nabla \cdot \vec{u})J$$

   This is the result we need in the proof of the Transport Theorem

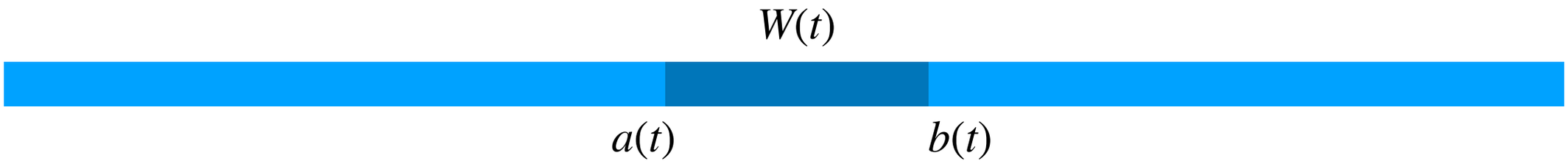
   where $\vec{u}$ is the flow velocity satisfying

$$\vec{u} = \frac{\partial \vec{x}}{\partial t}.$$

- Here J is the usual Jacobian from multi-variable calculus, or the determinant of the matrix of first partial derivatives of the vector valued mapping Phi.

- What makes the proof slippery is the way linear algebra and multi-variable calculus come together in it. There isn't anything conceptually difficult, but the combination of new notation and old ideas that were presented in isolation in our courses complicates things.

- Now recall the Conservation of Linear Momentum in 1D



The stress acts at ends

$$I(t) = \int_{W(t)} \rho(x,t)u(t,x)\ dx$$

- Assume that stress is the only force acting and find:

$$\frac{d}{dt}I(t) = \frac{d}{dt}\int_{W(t)} \rho(x,t)u(t,x)\ dx = \tau(b(t),t) - \tau(a(t),t) = \int_{W(t)} \frac{\partial \tau}{\partial x}\ dx$$

- Carry out the same mapping trick as for the Conservation of Mass to find

$$\int_{W(t)} \left[\frac{\partial u\rho}{\partial t} + u\frac{\partial u\rho}{\partial x} + u\rho\frac{\partial u}{\partial x} - \frac{\partial \tau}{\partial x}\right]\ dx = 0$$

- The Dubois Reymond Lemma then gives:

$$\frac{\partial u\rho}{\partial t} + u\frac{\partial u\rho}{\partial x} + u\rho\frac{\partial u}{\partial x} = \frac{\partial \tau}{\partial x}$$

The Reynolds Transport Theorem For Momentum

$$\frac{d\vec{Q}}{dt} = \iiint_{W(t)} \left[\frac{D\rho\vec{u}}{Dt} + \rho\vec{u}\nabla \cdot \vec{u} \right] dV$$

- But the rate of change of momentum isn't just zero, it is a sum of the applied forces.

- Some of these occur over the whole material volume (body forces). I write the body force as the negative gradient of a potential that I denote with a Π.

- Others occur only at the boundary (contact forces). These are written in terms of the traction, and then we apply Cauchy's theorem to express in terms of the stress tensor, and finally the Divergence Theorem to change the surface integral to a volume integral.

$$\iiint_{W(t)} -\nabla\Pi dV + \iint_{\partial W(t)} \vec{t}(\vec{x};\hat{n})dS$$

We will spend a whole slide deck on stress

$$\iiint_{W(t)} \left[-\nabla\Pi + \nabla \cdot \tau \right] dV$$

Putting it all together and using the Dubois-Reymond Lemma:

$$\iiint_{W(t)} \left[\frac{D\rho\vec{u}}{Dt} + \rho\vec{u}\nabla \cdot \vec{u} + \nabla\Pi - \nabla \cdot \tau \right] dV = 0$$

$$\frac{D\rho\vec{u}}{Dt} + \rho\vec{u}\nabla \cdot \vec{u} = -\nabla\Pi + \nabla \cdot \tau$$

When the density is constant, a simplification we will use a fair amount in this course the above simplifies to:

$$\rho_0 \frac{D\vec{u}}{Dt} = -\nabla\Pi + \nabla \cdot \tau$$

Notice how we used the conservation of mass to simplify the above

- It remains to relate the stress tensor, τ_{ij}, to the velocity.

- For this we follow the same pattern of algebra as in elasticity, but now it is the rate of strain we are talking about, and not the strain.

- This is because the basic variable of fluid mechanics is velocity, not displacement.

- As an aside this means the various material constants have different dimensions (i.e. units) in fluid mechanics, compared to elasticity.

$$e_{ij} = \frac{1}{2}\left(u_{i,j} + u_{j,i}\right), \mathbf{e} = \frac{1}{2}\left(\nabla\vec{u} + \nabla\vec{u}^T\right)$$

- Historically the stress is partitioned into an isotropic part (called pressure; the reason for the minus sign is a nice exercise) and a viscous part:

$$\tau_{ij} = -p\delta_{ij} + 2\mu e_{ij}, \tau = -p\mathbf{I} + 2\mu\mathbf{e}$$

- In the above I used incompressibility, or $\nabla \cdot \vec{u} = u_{i,i} = 0$. This is guaranteed when density is constant, as we showed a few slides ago.

- This saves us from talking about the bulk viscosity, a quantity that turns out to be difficult to measure.

- μ, the shear viscosity, is easy to measure with techniques going back to Newton.

- OK, finally we need to take the row-wise divergence of the stress tensor. The index notation is pretty handy in this context since we have:

$$\tau_{ij,j} = -p_{,i} + 2\mu e_{ij,j} = -p_{,i} + \mu u_{i,jj} + \mu u_{ij,i}$$

$$\boxed{\nabla \cdot \tau = -\nabla p + \mu \nabla^2 \vec{u}}$$

- We now have the closed from, Navier Stokes equations for a constant density fluid:

$$\boxed{\begin{gathered} \rho_0 \frac{D\vec{u}}{Dt} = -\nabla \Pi - \nabla p + \mu \nabla^2 \vec{u} \\ \nabla \cdot \vec{u} = 0 \end{gathered}}$$

- Euler argued that for many everyday fluids (e.g. water) shear viscosity was irrelevant and the corresponding set of Euler equations govern a constant density, inviscid fluid read:

$$\boxed{\rho_0 \frac{D\vec{u}}{Dt} = -\nabla \Pi - \nabla p, \nabla \cdot \vec{u} = 0}$$

- Euler's equations may seem like a pretty drastic reduction.

- However, measuring pressure has a long history:

- http://www.smar.com/en/technical-article/pressure-measurement-characteristics-technologies-and-trends

- And it is only a recent luxury, to focus on theories not motivated by measurement.

- It turns out Euler was partly right, but this needs a systematic inter comparison of the various terms in the equations.

Boundary Value Problems of Fluid Mechanics

- In the simplest context of constant density fluids it thus appears that Fluid Mechanics amounts to solving a set of four PDEs, one of which is linear and three of which have quadratic nonlinearities.

- The order of the highest derivative will depend on whether we want to consider the fluid as viscous or inviscid.

- To solve a problem we must impose boundary conditions.

- Physically we certainly understand that at a solid boundary the flux of fluid through the boundary must be zero. This is the so called no normal flow condition. I call it the "I am not Superman condition" and reads $\vec{u} \cdot \hat{n} = 0$

- This condition holds regardless of whether we wish to include or ignore viscous effects (for Euler and Navier Stokes equations).

- For Navier-Stokes equations the non-zero shear stresses also tell us that the fluid must not be moving in a tangential direction to the solid boundary.

Boundary Value Problems of Fluid Mechanics

- For Navier-Stokes equations the non-zero shear stresses also tell us that the fluid must not be moving in a tangential direction to the solid boundary, $\vec{u} \cdot \hat{t} = 0$, where $\hat{n} \cdot \hat{t} = 0$.

- Together these two conditions tell us that the velocity of a viscous fluid at a solid boundary is zero, $\vec{u} = 0$, the so-called no slip boundary condition.

- Of course if we wish to apply a force to a fluid we must use Cauchy's theorem, to get $\tau_{ij} n_j = -p n_i + 2\mu e_{ij} n_j = F_i^{(applied)}$

- This shows that for inviscid fluid the pressure must match the applied force.

- For viscous fluids things are considerably more complicated.

- In modern fluid dynamics solutions are usually arrived at numerically and imposing boundary conditions correctly is often the most challenging part of numerical methods.

Non-dimensionalization Overview

- The Navier Stokes equations describe phenomena on scales from the sub millimeter to thousands of kilometers.
- It is thus an obvious mathematical goal to put them into a form that is dimensionless.
- This can be tricky in complex situations, but the basics are fairly clear; as we are about to demonstrate.
- The key to non-dimensionalization is to choose representative values of length (L), velocity (U) and so on. Thus for flow over an office building U might be 10 meters a second, and L might be 30 meters.

Non-dimensionalization of the momentum equations

$$\rho_0[u_t + uu_x + wu_z] = -p_x + \mu\nabla^2 u$$

$$\vec{u} = U\tilde{u}\ ,\ (x,z) = L(\tilde{x},\tilde{z})\ ,\ p = \rho_0 U^2\tilde{p}\ ,\ t = \frac{L}{U}\tilde{t}$$

$$\rho_0\frac{U^2}{L}[\tilde{u}_{\tilde{t}} + \tilde{u}\tilde{u}_{\tilde{x}} + \tilde{w}\tilde{u}_{\tilde{z}}] = \rho_0\frac{U^2}{L}(-\tilde{p}_{\tilde{x}}) + \frac{\mu U}{L^2}\tilde{\nabla}^2\tilde{u}$$

divide through, let $Re = \frac{UL}{\nu}$ and drop tildes.

$$u_t + uu_x + wu_z = -p_x + \frac{1}{Re}\nabla^2 u$$

$Re \to \infty$ means we get Euler equations

For the office building example on the last slide

$Re = Ul/\nu \approx 10 \times 30/1 \times 10^{-6} = 3 \times 10^7$

So very close to the limit mentioned above

Non-dimensionalization

- The Reynolds number is the ratio between a typical nonlinear (physicist often say “inertia”) term and a typical viscous term.
- High Re means inertia is more important than viscosity, or in other words dissipation doesn’t really matter.
- Low Re means the viscous terms matter more. This could be because the viscosity is high (like say for honey) or the lengths are tiny (like say for a zoo-plankton specimen)